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Studierichting: Informatica

Oefeningenreeks 4A nr. 38.6

$$\begin{aligned}\int \frac{5x-7}{x^2+3x+3} dx &= \int \frac{\frac{5}{2}(2x+3) - \frac{29}{2}}{x^2+3x+3} dx = \frac{5}{2} \int \frac{d(x^2+3x+3)}{x^2+3x+3} - \frac{29}{2} \int \frac{1}{x^2+3x+3} dx \\&= \frac{5}{2} \ln|x^2+3x+3| - \frac{29}{2} \int \frac{1}{\left(x+\frac{3}{2}\right)^2 + \frac{3}{4}} d\left(x+\frac{3}{2}\right) \\&= \frac{5}{2} \ln|x^2+3x+3| - \frac{29}{\sqrt{3}} \operatorname{Bgtan}\left(\frac{2x+3}{\sqrt{3}}\right) + c\end{aligned}$$

References